

MATH 301 - Project 1

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October 1, 2025

1 Abstract

This project investigates the problem of achieving a safe and controlled landing of a quadrotor in the presence of aerodynamic ground effects. The quadrotor's vertical dynamics are modeled by a second-order nonlinear differential equation, where thrust is amplified near the ground due to airflow interactions. The primary objective is to design a feedback controller that enables the quadrotor to converge reliably to a small target height above the surface without crashing. To address this, the nonlinear dynamics were linearized around a chosen equilibrium point, and a state-space representation was developed. A linear state feedback law of the form $u(t) = -k_1x_1(t) - k_2x_2(t)$ was implemented in MATLAB, where the gains k_1, k_2 were tuned through repeated simulations. Performance was evaluated using two metrics: (1) the percentage of successful landings over 100 trials and (2) the average settling time within 10% of the target height. Results demonstrate that appropriately tuned feedback gains achieve a high rate of successful landings with reduced convergence time, highlighting the importance of gain selection. This study provides a foundation for more advanced quadrotor control strategies and extensions to complex landing scenarios.

2 Introduction and Problem Description

Quadrotors and other small-scale unmanned aerial vehicles (UAVs) have become widely used in applications ranging from aerial photography to environmental monitoring and package delivery. One of the critical challenges for safe and reliable operation is autonomous landing, where the vehicle must reduce its height and velocity in a controlled way to avoid hard impacts with the ground. This task becomes more complex due to ground effects: aerodynamic disturbances that arise when the rotor airflow interacts with the surface, amplifying thrust and altering stability. As the vehicle approaches the ground, these effects must be accounted for in order to design a robust controller.

In this project, the focus is on modeling the vertical dynamics of a quadrotor and designing a feedback controller that ensures a safe descent. The system is described by a nonlinear differential equation in which thrust is modified by a height-dependent amplification factor. To address the landing problem, the nonlinear model is linearized about a chosen equilibrium point just above the ground, and a state feedback controller is implemented. By tuning the feedback gains, the quadrotor can be made to converge quickly and reliably to a small hover height without overshooting or crashing.

Practical applications of such a landing controller include search-and-rescue drones, where precise ground approaches are necessary in constrained environments, and logistics drones, where repeated landings must be safe and efficient. More broadly, the techniques explored here form the foundation for advanced UAV control methods that deal with nonlinearities and environmental disturbances.

The main results of this project show that carefully selected feedback gains yield a high percentage of successful landings and relatively short convergence times. In contrast, poorly tuned gains can lead to instability or crashes. The findings highlight both the importance of gain tuning and the usefulness of linearization as a tool for simplifying nonlinear dynamics in control system design.

3 Linearization and Feedback

The verticle motion of quadrotor is described by:

$$m\ddot{z} = -mg + (1 + F_G(z))T \quad F_G(z) = \frac{c_1}{z^2 + c_2}$$

where z is the altitude, T is thrust, $F_G(z)$ is the ground effects.

We want the quadrotor to hover steadily just above the ground before landing, so we pick a small altitude, $z_0 = 0.001m$, which means we can get velocity and accerlation as:

$$\dot{z} = 0 \quad \ddot{z} = 0$$

At equalibrium, the thrust balances gravity with ground effect is:

$$T_0 = \frac{mg}{1 + F_G(z_0)}$$

which for the state-space form $\dot{x} = Ax + Bu$, we have:

$$A = \begin{bmatrix} 0 & 1 \\ \frac{T_0}{m} F'_G(z_0) & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ \frac{1 + F_G(z_0)}{m} \end{bmatrix}$$

In our case, we set fitting the parameters in the ground effect model to the experimental data gives approximately $c_1 = 0.008$ and $c_2 = 0.006$, which means:

$$A = \begin{bmatrix} 0 & 1 \\ -1.87 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ \frac{2.333}{m} \end{bmatrix}$$

For the state feedback, state feedback means using measurements of the system's states. In here, height error x_1 and verticle velocity x_2 , to adjust the control input. We have:

$$u(t) = -k_1 x_1(t) - k_2 x_2(t)$$

In this equation, k_1 is the corrects for altitude error, and k_2 is the damps the velocity. With k_1 and k_2 , we can make the quadrotor smoothly converge to a small hover height near the gound, which make the quadrotor ready for a safe landing.

4 Analysis of Gains

I began by selecting small values of k_1 and k_2 to ensure stability and avoid oscillations. For k_1 , it is the correction of the altitude error, which means larger values make the quadrotor converge faster to the desired altitude. For k_2 , it is velocity gain, which means larger values add damping, preventing overshoot or oscillations near the ground. In other words, k_1 is for the responsiveness, and k_2 is for the smoothness.

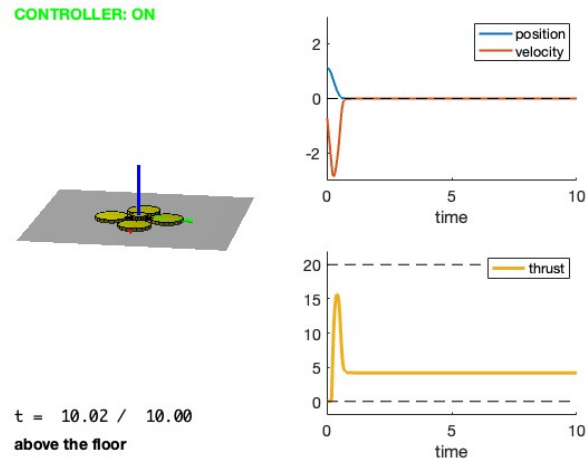


Figure 1: Simulation when $k_1 = 30$ and $k_2 = 8.995$

For each gain set, we can get two related numbers: Success rate and the average settling time. Here is the list of the result that I got for repeat 100 times:

k_1	k_2	Success Rate	Avg. Settling Time
10	5	0%	NaN
15	10	100%	4.352s
20	15	100%	5.469s
25	20	100%	6.023s
30	25	100%	6.608s
15	5	0%	NaN
20	10	100%	3.377s
25	10	100%	2.737s
30	10	100%	2.281s
30	8	0%	NaN
30	9	90%	1.326s
30	8.5	0%	NaN
30	8.8	31%	1.078s
30	8.7	11%	1.082s
30	8.6	1%	1.360s
30	8.9	67%	1.232s
30	8.95	83%	1.273s
30	8.98	87%	1.205s
30	8.99	93%	1.341s
30	8.995	95%	1.395s

From those results, several trends can be seen:

- (1) Small gains failed completely, as the thrust corrections were too weak to stabilize the system.
- (2) Moderate gains usually could achieve 100%, with the settling time is around 2-3s, which consider both efficiency and the reliability.
- (3) Large gains still achieve 100%, but settling times grew longer, which indicate over-damped, sluggish landings.
- (4) When we adjust near instability, produced a very short amount of settling times, but it could significantly lower the success rates.

All in all, there is a clear trade-off between speed and reliability. Moderate could balance success rate and settling time, and aggressive one could short time but increase the risk of fail.

5 Conclusion and Future Directions

In this project, I tested a state-feedback controller to stabilize a quadrotor during landing in the presence of ground effect. By linearizing the nonlinear dynamics around a small hover height, I implemented a feedback law based on altitude error and velocity. Through systematic gain tuning, I found that moderate values of k_1 (responsiveness) and k_2 (damping) produced the most reliable performance, achieving 100% success rates with average settling times of 2–4 seconds. Aggressive tuning reduced settling time to near 1 second but at the cost of significantly lower success rates,

highlighting the trade-off between speed and robustness.

Future work could extend this controller in several ways. We could testing the controller in more realistic scenarios, which could leading to more gains that finely adjust the simulation to make it more close to the real world, like adding wind disturbances or sensor noise. This could make the results more generalizable to the real-world quadrotor applications.